Ebad Rehman

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Summary

Software Engineering student with strong proficiency in Full Stack development (React, Next.js), Embedded Systems (C/C++, STM32), and Cloud Infrastructure (AWS). Passionate about engineering scalable web applications and optimizing low-level firmware, actively seeking internship opportunities to contribute to complex technical challenges.

Education

University of Calgary — **Graduation: May 2027**
B.Sc. in Software Engineering — *Calgary, AB*
- **Certification:** AWS Solutions Architect - Associate

Technical Skills

Languages: Python, Java, C, C++, JavaScript, TypeScript, SQL, Bash (Linux)

Cloud & Infrastructure: AWS (Lambda, S3, DynamoDB, EC2), Serverless Architecture, ETL Pipelines

Frameworks: React, Next.js, Express.js, Flask, LangChain, LangGraph, Spring Boot, Swing (Java)

Tools: Git, MySQL, NoSQL (MongoDB), n8n

Experience

TechQuest — **June 2025 – August 2025**
Software Engineering Intern — *Calgary, AB*
- Reduced manual workflow configuration time by **60%** in multi-agent AI systems by implementing **LangChain** and **LangGraph**, automating **5+ pipelines** via **n8n**.
- Reduced infrastructure costs by **30%** by architecting a **serverless AWS infrastructure** (**Lambda**, **S3**, **DynamoDB**) to process **1,000+ daily IoT events**.
- Delivered **3 production AI agents** (Weather, EV Config, Search) by collaborating with an **8-engineer team** on system architecture reviews and **API integrations**.

Freelance — **December 2023 – Present**
Full Stack Developer — *Calgary, AB*
- Maximized **SEO ranking** and load speeds by architecting high-performance web platforms with **Next.js (React)** and **TypeScript**, implementing **Static Site Generation (SSG)**.
- Implemented a routing-based localization system that dynamically adapted layout direction (**RTL/LTR**) for English–Arabic translations across mobile and desktop platforms.
- Sustained **3,000+ monthly active users** by integrating **Google Analytics** to drive continuous performance optimizations and content strategy.

Projects

FightMetrics | *AI-Powered Athletic Performance Tracker* — **Python, Scikit-Learn, Pandas, React**
- Identified ineffective workout patterns by building a recommendation engine using **K-Means clustering (Scikit-Learn)**, optimizing training schedules for athletes.
- Enabled processing of **50+ daily data points** per user by engineering an injury-prevention system with **Pandas rolling windows** to flag high-risk **"Acute:Chronic" ratios**.
- Deployed a **real-time burnout prediction pipeline** utilizing **Socket.IO** to push live injury-risk alerts to a **React/TypeScript dashboard**.

STM32 Kinematics Engine | *Real-Time Signal Processing* — **C/C++, Bare-metal, DMA, Interrupts**
- Engineered a **bare-metal embedded system** on STM32 (Cortex-M4) using **C/C++**, bypassing standard HAL libraries to directly manipulate hardware registers for **low-latency I/O**.
- Achieved **10+ kHz ADC sampling** with less than **5% CPU utilization** by implementing a **Zero-Copy** data pipeline using **Direct Memory Access (DMA)** and **Circular Buffers**.
- Guaranteed **deterministic execution** for real-time kinematics analysis by designing a **hardware interrupt-driven scheduler** to manage concurrent tasks.

Flight Reservation System | *Object Oriented Java Application* — **Java, Swing, SQL**
- Reduced codebase size by **35-50%** and boosted maintainability by applying **design patterns** and **SOLID principles** across **35+ classes** in a robust **Java** system.
- Designed a **relational database** with **8+ SQL tables** for managing complex relationships between flights, passengers, and bookings.
- Secured application access by developing role-based user interfaces (Customer, Agent, Admin) in the **Swing framework**.